Gerasimos Gerogiannis

Research Interests

- Computer architecture and high-performance computing.
- Accelerator-centric heterogeneous architectures for machine learning, graph analytics, and scientific workloads.
- Machine learning and its application to systems and computer architecture.

Professional Appointments

University of California, Irvine — Oct. 2027 -
Assistant Professor, Department of Computer Science — *Irvine, CA, USA*

Google — Aug. 2026 - Aug. 2027
Visiting Faculty Researcher — *Sunnyvale, CA, USA*

Education

University of Illinois at Urbana-Champaign (UIUC) — Aug. 2021 - Aug. 2026
PhD in Electrical and Computer Engineering — *Urbana, IL, USA*

- Thesis: *'Coupling, Scaling, and Optimizing Accelerator-Centric Systems'*
- Advisor: Prof. Josep Torrellas
- GPA: 4.0/4.0

University of Patras — Sep. 2016 - June 2021
Diploma in Electrical and Computer Engineering (BSc + MSc) — *Patras, Greece*

- Thesis: *'Reinforcement Learning for Task Offloading in Next Generation Networks: Algorithms and Hardware Acceleration'*
- Thesis Supervisor: Prof. Alexios Birbas
- GPA: 9.83/10.00 (first in class)

Work Experience

Intel Corporation — Jan. 2024 - Dec. 2025
CPU Architecture & Performance Intern — *(remote) Champaign, IL, USA*

- CPU architecture redesign for emerging machine learning applications such as large language models.
- Research conducted during the internship led to publications in MICRO'25 and ISCA'26 and 4 filed U.S. patents.

Intel Corporation — May 2023 - Aug. 2023
Accelerator Architecture & Performance Intern — *(remote) Champaign, IL, USA*

- Heterogeneous accelerator architectures to accelerate sparse-dense matrix multiplication.
- Research conducted during the internship led to a publication in HPCA'24.

Intel Corporation — May 2022 - Aug. 2022
Accelerator Architecture & Performance Intern — *(remote) Champaign, IL, USA*

- Synergies between hardware and software to maximize performance on the Intel PIUMA accelerator.
- Research conducted during the internship led to a publication in ISPASS'23.

i-acoma group, University of Illinois at Urbana-Champaign — Aug. 2021 - Aug. 2026
Research Assistant — *Champaign, IL, USA*

Awards and Honors

MICRO PhD Forum IEEE TCuARCH, 2025

- Selected to present my PhD research at the PhD Forum of the International Symposium on Microarchitecture.

Wen-mei W. Hwu Award UIUC, ECE Department, 2025

- For demonstrating research expertise in the area of Computer Engineering.

Mavis Future Faculty Fellowship UIUC, Grainger College of Engineering, 2025

- Awarded to facilitate the training of the next generation of engineering faculty.

Rambus Computer Engineering Fellowship UIUC, ECE Department, 2024

- For demonstrating outstanding research performance in the area of Computer Engineering.

IEEE Micro Top Picks from Computer Architecture Conferences Top Pick IEEE, 2024

- Awarded to the 12 most significant papers in computer architecture published in the previous year based on novelty and potential for long-term impact. Top Pick awarded for paper: 'Micro-Armed Bandit: Lightweight & Reusable Reinforcement Learning for Microarchitecture Decision-Making'.

IEEE Micro Top Picks from Computer Architecture Conferences Honorable Mention IEEE, 2024

- Awarded to the 12 most significant papers in computer architecture published in the previous year based on novelty and potential for long-term impact. Honorable Mention awarded for paper: 'SPADE: A Flexible and Scalable Accelerator for SpMM and SDDMM'.

Conference Travel Grants for ISCA/MICRO/HPCA/ASPLOS IEEE/ACM, 2023-2025

Student Excellence Award University of Patras, Greece, 2021

- For the highest GPA among all 2020-2021 graduates of the ECE Department of the University of Patras.

Excellence Award State Scholarships Foundation (IKY), Greece, 2021

- For the highest GPA among all 2020-2021 graduates of the ECE Department of the University of Patras.

Skouras Foundation Award Skouras Foundation, Greece, 2020

- For the highest GPA among all undergraduate students in the ECE Department of the University of Patras.

Freshman Excellence Award University of Patras, Greece, 2016

- For the highest grade in the National University Entrance Exams (Panhellenic Exams) among the first-year students in the ECE Department of the University of Patras.

Distinctions in National High-school Student Competitions 2013-2016

- Distinctions in National Student Physics, Chemistry, and Math Competitions. Auxiliary member of the national team for the International Physics Olympiad.

Conference Publications

[C14] *'ATX: Accelerator Task Extensions'*
Gerasimos Gerogiannis, Stijn Eyerman, Josep Torrellas, and Wim Heirman
In Proceedings of the International Symposium on Computer Architecture (ISCA), July 2026.

[C13] *'Dorado: Clustered Hardware Cache Coherence for 1,000+ Cores'*
Jovan Stojkovic, Abraham Farrell, **Gerasimos Gerogiannis**, Zhangxiaowen Gong, Christopher J. Hughes, and Josep Torrellas
In Proceedings of the International Symposium on Computer Architecture (ISCA), July 2026.

[C12] *'GRANII: Selection and Ordering of Primitives in GRAph Neural Networks using Input Inspection'*
Damitha Lenadora, Vimarsh Sathia, **Gerasimos Gerogiannis**, Serif Yesil, Josep Torrellas, and Charith Mendis
In Proceedings of the International Symposium on Code Generation and Optimization (CGO), February 2026.

[C11] *'NetSparse: In-Network Acceleration of Distributed Sparse Kernels'*
Gerasimos Gerogiannis, Dimitrios Merkouriadis, Charles Block, Annus Zulfiqar, Filippos Tofalos, Muhammad Shahbaz, and Josep Torrellas
In Proceedings of the International Symposium on Microarchitecture (MICRO), October 2025.

[C10] *'DECA: A Near-Core LLM Decompression Accelerator Grounded on a 3D Roofline Model'*
Gerasimos Gerogiannis, Stijn Eyerman, Evangelos Georganas, Wim Heirman, and Josep Torrellas
In Proceedings of the International Symposium on Microarchitecture (MICRO), October 2025.

[C9] *'Micro-MAMA: Multi-Agent Reinforcement Learning for Multicore Prefetching'*
Charles Block, **Gerasimos Gerogiannis**, and Josep Torrellas
In Proceedings of the International Symposium on Microarchitecture (MICRO), October 2025.

[C8] *'COGNATE: Learning-Based Acceleration of Sparse Tensor Programs on Emerging Hardware'*
Chamika Sudusinghe, **Gerasimos Gerogiannis**, Damitha Lenadora, Charles Block, Josep Torrellas, and Charith Mendis
In Proceedings of the International Conference on Machine Learning (ICML), July 2025.

[C7] *'MeshSlice: Efficient 2D Tensor Parallelism for Distributed DNN Training'*
Hyoungwook Nam, **Gerasimos Gerogiannis**, and Josep Torrellas
In Proceedings of the International Symposium on Computer Architecture (ISCA), June 2025.

[C6] *'Distributed-Memory Parallel Algorithms for Sparse Matrix and Sparse Tall-and-Skinny Matrix Multiplication'*
Isuru Ranawaka, Md Taufique Hussain, Charles Block, **Gerasimos Gerogiannis**, Josep Torrellas, and Ariful Azad
In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC), November 2024.

[C5] *'Two-Face: Combining Collective and One-Sided Communication for Efficient Distributed SpMM'*
Charles Block*, **Gerasimos Gerogiannis***, Charith Mendis, Ariful Azad, and Josep Torrellas
*Co-first authors, order is alphabetical.
In Proceedings of the International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), April 2024.

[C4] *'HotTiles: Accelerating SpMM with Heterogeneous Accelerator Architectures'*
Gerasimos Gerogiannis, Sriram Aananthakrishnan, Josep Torrellas, and Ibrahim Hur
In Proceedings of the International Symposium on High Performance Computer Architecture (HPCA), March 2024.

[C3] *'Micro-Armed Bandit: Lightweight & Reusable Reinforcement Learning for Microarchitecture Decision-Making'*
Gerasimos Gerogiannis and Josep Torrellas
In Proceedings of the International Symposium on Microarchitecture (MICRO), October 2023.
Selected as a Top Pick in IEEE Micro Top Picks from Computer Architecture Conferences.

[C2] *'SPADE: A Flexible and Scalable Accelerator for SpMM and SDDMM'*
Gerasimos Gerogiannis, Serif Yesil, Damitha Lenadora, Dingyuan Cao, Charith Mendis, and Josep Torrellas
In Proceedings of the International Symposium on Computer Architecture (ISCA), June 2023.
Selected as an Honorable Mention in IEEE Micro Top Picks from Computer Architecture Conferences.

[C1] *'Characterizing the Scalability of Graph Convolutional Networks on Intel® PIUMA'*
Matthew Adiletta, Jesmin Jahan Tithi, Emmanouil-Ioannis Farsarakis, **Gerasimos Gerogiannis**, Robert Adolf, Robert Benke, Sidharth Kashyap, Samuel Hsia, Kartik Lakhotia, Fabrizio Petrini, Gu-Yeon Wei, and David Brooks
In Proceedings of the International Symposium on Performance Analysis of Systems and Software (ISPASS), April 2023.

Journal Publications

[J2] *'Practical Online Reinforcement Learning for Microprocessors with Micro-Armed Bandit'*
Gerasimos Gerogiannis and Josep Torrellas
IEEE Micro Magazine, Top Picks in Computer Architecture Special Issue, July-August 2024.

[J1] *'Deep Reinforcement Learning Acceleration for Real-Time Edge Computing Mixed Integer Programming Problems'*
Gerasimos Gerogiannis, Michael Birbas, Aimilios Leftheriotis, Eleftherios Mylonas, Nikolaos Tzanis, and Alexios Birbas
IEEE Access, January 2022.

Workshop Publications (with proceedings)

[W2] *'Performance-Driven Composite Prefetching with Bandits'*
Charles Block, Pedro Palacios Almendros, Abraham Farrell, **Gerasimos Gerogiannis**, and Josep Torrellas
In Proceedings of the 4th Data Prefetching Championship (DPC4) at HPCA 2026.

[W1] *'Automated Data Selection for Efficient Cost Model Training to Optimize Sparse Matrix Kernels on Emerging Hardware Accelerators'*
Chamika Sudusinghe, **Gerasimos Gerogiannis**, Damitha Lenadora, Charles Block, Josep Torrellas, and Charith Mendis
In Proceedings of the Exploration in AI Today Workshop at ICML 2025.

U.S. patents (filed)

[P4] *'Unified Transfer Engine for Near-Core Compute Accelerators'*
Gerasimos Gerogiannis, Stijn Eyerman, and Wim Heirman
U.S. Patent App. 19/342,503 , January 2026

[P3] *'Speculative Invocation of Accelerators in Out-of-Order Core Pipelines'*
Gerasimos Gerogiannis, Stijn Eyerman, and Wim Heirman
U.S. Patent App. 19/342,390 , January 2026

[P2] *'Efficient On-chip Memory Bandwidth Configurations and Distribution for Artificial Intelligence (AI) Workloads'*
Wim Heirman, Stijn Eyerman, and **Gerasimos Gerogiannis**

[P1] *'Methods and Apparatus for a Machine Learning Model Decompression Accelerator'*
Gerasimos Gerogiannis, Stijn Eyerman, Wim Heirman, and Evangelos Georganas
U.S. Patent App. 18/927,638 , November 2024.

Talks

[T13] *'ATX: Accelerator Task Extensions'*
International Symposium on Computer Architecture (ISCA), Raleigh, NC, June 2026.

[T12] *'Heterogeneity without the Headache: Architecting Accelerator-Centric Computing Systems'*
Imperial College London, April 2026
The University of Texas at Austin, March 2026
Arizona State University, March 2026
University of Pennsylvania, March 2026
University of Colorado Boulder, March 2026
University of California, Irvine, March 2026
University of Massachusetts Amherst, March 2026
University of Minnesota, February 2026
Indiana University, February 2026
University of Waterloo, January 2026

[T11] *'Towards Accelerator-Centric Computing'*
9th Computing Systems Research Day, National Technical University of Athens, Greece, January 2026

[T10] *'Accelerator-Centric Computing'*
PhD Forum at the 58th International Symposium on Microarchitecture, Seoul, South Korea, October 2025.

[T9] *'NetSparse: In-Network Acceleration of Distributed Sparse Kernels'*
International Symposium on Microarchitecture (MICRO), Seoul, South Korea, October 2025.

[T8] *'DECA: A Near-Core LLM Decompression Accelerator Grounded on a 3D Roofline Model'*
International Symposium on Microarchitecture (MICRO), Seoul, South Korea, October 2025.

[T7] *'Sparsity at Scale: Towards Efficient Distributed Sparse Accelerators'*
SRC TECHCON, Austin, TX, September 2024.

[T6] *'Two-Face: Combining Collective and One-Sided Communication for Efficient Distributed SpMM'*
International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), San Diego, CA, April 2024.

[T5] *'HotTiles: Accelerating SpMM with Heterogeneous Accelerator Architectures'*
International Symposium on High Performance Computer Architecture (HPCA), Edinburgh, United Kingdom, March 2024.

[T4] *'Micro-Armed Bandit: Lightweight & Reusable Reinforcement Learning for Microarchitecture Decision-Making'*
Intel Archfest, Online, May 2024.
International Symposium on Microarchitecture (MICRO), Toronto, Canada, October 2023.

[T3] *'Domain-Specific Hardware and Software for Mixed Sparse-Dense Algebra at Scale'*
ACE Center for Evolvable Computing Theme Meeting, Online, August 2023.

[T2] *'SPADE: A Flexible and Scalable Accelerator for SpMM and SDDMM'*
International Symposium on Computer Architecture (ISCA), Orlando, FL, June 2023.

[T1] *'Accelerators for Irregular Applications'*
IBM-Illinois Discovery Accelerator Institute Annual Meeting, Urbana, IL, March 2023.

Selected Projects

MiniSPADE 2022-2023

- Designed the microarchitecture and taped out a simplified ASIC chip prototype of the accelerator described in our ISCA'23 paper using TSMC 65nm technology.

Teaching Experience

Teaching Assistant ECE Department, UIUC, Spring 2026

- Teaching assistant for ECE 445 Senior Design Project.

Tutor Prosimo Education Center, Patras, Greece, 2020 – 2021

Private Tutor Patras, Greece, 2016 – 2020

Mentoring Experience

Graduate Student Mentoring UIUC, 2022 – present

- *Dimitrios Merkouriadis*, PhD student at UIUC: In-Network Acceleration for Sparse Kernels, 2024 - present.
- *Filippos Tofalos*, PhD student at UIUC: In-Network Acceleration for Sparse Kernels, 2024 - 2025.
- *Chamika Sudusinghe*, PhD student at UIUC: Accelerator Performance Modeling and Tuning, 2023 - present.
- *Charles Block*, PhD student at UIUC: Algorithms for Distributed Sparse Kernels, 2022 - 2024.

Grant Writing Experience

NSF 2023 – 2028
PPoSS: LARGE: General-Purpose Scalable Technologies for Fundamental Graph Problems *Amount: $5,000,000*

- Lead PI: Josep Torrellas; number of PIs: 10.
- Assisted in the preparation and writing of the NSF Grant proposal, focusing on hardware support for scalable graph algorithms and on the hardware-software interaction.

Professional Memberships

IEEE July 2026 – present
Member

ACM May 2023 – present
Member

Service

Conference Program Committee Member
IISWC 2026
MICRO 2026
ASAP 2026
ACM SoCC 2026

Journal Reviewer
ACM Transactions on Architecture and Code Optimization (TACO)
IEEE Computer Architecture Letters (CAL)
IEEE Transactions on Computers (TC)

Technical Skills

Programming Languages: C/C++, Python, MATLAB, Shell/Bash scripting, Perl, VHDL, Verilog.
Parallel Programming: OpenMP, MPI, POSIX Threads, CUDA.
Frameworks/Libraries: Vivado, Vivado HLS, Cadence EDA tools, TensorFlow, PyTorch, Deep Graph Library.
Microarchitectural Simulators and Tools: SST, Sniper, gem5, ChampSim, Accel-Sim, DRAMSim, CACTI, McPAT.
Development Tools: Git, CMake.